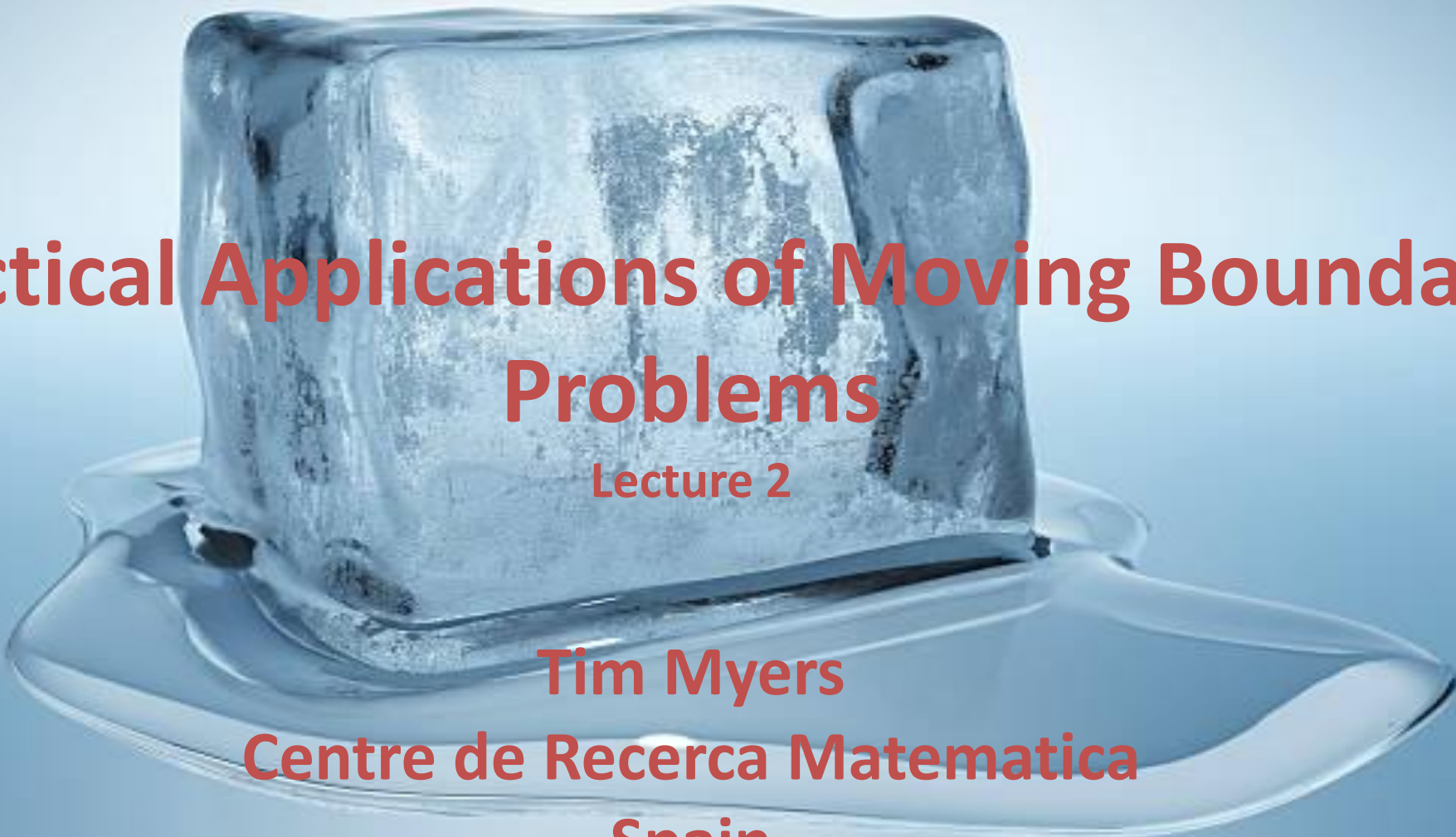




Practical Applications of Moving Boundary Problems

Lecture 2

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Steady-State

$$\frac{\partial(\Delta T \hat{T})}{\partial(\tau \hat{t})} = \alpha \frac{\partial^2(\Delta T \hat{T})}{\partial(\mathcal{L} \hat{x})^2} \Rightarrow \frac{\partial \hat{T}}{\partial \hat{t}} = \frac{\alpha \tau}{\mathcal{L}^2} \frac{\partial^2 \hat{T}}{\partial \hat{x}^2}$$

Now drop hat notation - should be easy to distinguish.

Subject to $T(0, t) = 0$, $T(L, t) = 1$

Steady-state

$$\frac{\partial^2 T}{\partial x^2} = 0 \Rightarrow T = x$$

Doesn't matter what value α takes

No IC to impose

Approach to Steady-State

Now look at large time behaviour (i.e. before we reach the steady-state).

Choose τ sufficiently large that $\frac{\mathcal{L}^2}{\alpha\tau} = \epsilon \ll 1$

$$\frac{\partial^2 T}{\partial x^2} = \epsilon \frac{\partial T}{\partial t}$$

Intuitively, for small ϵ expect $T \approx x$, but what is the error?

Perturbation Methods

These show how to deal
with small parameters
and indicate the size of
the error

Begin with a simple example ...

Consider the quadratic equation

$$x^2 - 2\epsilon x - 1 = 0 \quad \epsilon \ll 1$$

Exact solution

$$x = \frac{2\epsilon \pm \sqrt{4\epsilon^2 + 4}}{2} = \epsilon \pm \sqrt{1 + \epsilon^2}$$

Since $\epsilon \ll 1$

$$\begin{aligned} x &\approx \epsilon \pm \left(1 + \frac{\epsilon^2}{2} - \frac{\epsilon^4}{8} + \dots \right) \\ &= \pm 1 + \epsilon \pm \frac{\epsilon^2}{2} \mp \frac{\epsilon^4}{8} + \mathcal{O}(\epsilon^6) \end{aligned}$$

Perturbation Methods

$$x = \pm 1 + \epsilon \pm \frac{\epsilon^2}{2} \mp \frac{\epsilon^4}{8} + \mathcal{O}(\epsilon^6)$$

Take $\epsilon = 0.1$ and the positive root we find

$$x_0 = 1 \quad x_1 = 1.1 \quad x_2 = 1.105 \quad x_3 = 1.1049875$$

Exact $x = 1.10498756$, i.e. in 3 terms we have 3 decimal place accuracy

This improves as $\epsilon \rightarrow 0$

Perturbation Methods

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What if we try a series solution from the start?

Let $x = x_0 + \epsilon x_1 + \epsilon^2 x_2 + \mathcal{O}(\epsilon^3)$

Substitute into the original equation

$$(x_0 + \epsilon x_1 + \epsilon^2 x_2)^2 - 2\epsilon(x_0 + \epsilon x_1 + \epsilon^2 x_2) - 1 = 0$$

Gather together terms of the same order

$$x_0^2 - 1 + \epsilon(2x_0x_1 - 2x_0) + \epsilon^2(x_1^2 + 2x_0x_2 - 2x_1) + \mathcal{O}(\epsilon^3) = 0$$

Solve at each order of ϵ

$$\epsilon^0 : x_0^2 - 1 = 0 \quad x_0 = \pm 1 \quad = \text{solution when } \epsilon=0$$

$$\epsilon^1 : 2x_0x_1 - 2x_0 = 0 \quad x_1 = 1 \quad \text{correction terms}$$

$$\epsilon^2 : x_1^2 + 2x_0x_2 - 2x_1 = 0 \quad x_2 = \pm 1/2$$

Note, x_1 depends on x_0 , x_2 on x_1, x_0 etc

Hence

$$x = \pm 1 + \epsilon \pm \frac{\epsilon^2}{2} + \mathcal{O}(\epsilon^3) = \text{expansion of the exact solution}$$

What is Perturbation Theory

In general we would not use perturbation techniques to solve a quadratic equation.

Wikipedia: Perturbation Theory

Perturbation theory comprises mathematical methods that are used to find an approximate solution to a problem which cannot be solved exactly, by starting from the exact solution of a related problem (in the previous example we first solved $x^2 = 1$). Perturbation theory is applicable if the problem at hand can be formulated by adding a "small" term to the mathematical description of the exactly solvable problem.

Note, the terms *perturbation* and *asymptotic* expansion are often used interchangeably (see Howison P177)

What is Perturbation Theory

Bender & Orszag

Perturbation theory is a large collection of iterative methods for obtaining approximate solutions to problems involving a small parameter ϵ . These methods are so powerful that sometimes it is advisable to introduce a small parameter temporarily into a difficult problem ... finally to set $\epsilon = 1$ to recover the original problem

The thematic approach of perturbation theory is to decompose a tough problem into an infinite number of relatively easy ones.

So, once we understand how to perturb a simple problem, we may start to apply the technique to more difficult ones ...

Order notation

First a small aside -

The notation $\mathcal{O}()$ may describe the error term in an approximation to a mathematical function. The most significant terms are written explicitly and then the less significant terms are summarized in a single $\mathcal{O}()$ term. For example,

$$\exp(x) = 1 + x + \frac{x^2}{2} + \mathcal{O}(x^3) \quad \text{as } x \rightarrow 0$$

Indicates that

$$\left| \exp(x) - \left(1 + x + \frac{x^2}{2}\right) \right| < C|x^3| \quad \text{as } x \rightarrow 0$$

A small example

Consider the differential equation with a small parameter

$$\frac{du}{dt} + \epsilon u = t \quad u(0) = 1$$

Homogeneous solution $u = Ae^{mt}$, $m + \epsilon = 0 \Rightarrow u = Ae^{-\epsilon t}$

Particular integral, try $u = a + bt$

$$b + \epsilon(a + bt) = t \quad a = -b/\epsilon \quad b = 1/\epsilon$$

Exact solution

$$\begin{aligned} u &= Ae^{-\epsilon t} + \frac{t}{\epsilon} - \frac{1}{\epsilon^2} \quad u(0) = 1 \\ &= \left(1 + \frac{1}{\epsilon^2}\right) e^{-\epsilon t} + \frac{t}{\epsilon} - \frac{1}{\epsilon^2} \end{aligned}$$

A small example

$$u = \left(1 + \frac{1}{\epsilon^2}\right) e^{-\epsilon t} + \frac{t}{\epsilon} - \frac{1}{\epsilon^2}$$

Expand exponential for small ϵt

$$\begin{aligned} u &\approx \left(1 + \frac{1}{\epsilon^2}\right) \left(1 - \epsilon t + \epsilon^2 \frac{t^2}{2} - \epsilon^3 \frac{t^3}{6} \dots\right) + \frac{t}{\epsilon} - \frac{1}{\epsilon^2} \\ &= 1 + \frac{t^2}{2} - \epsilon \left(t + \frac{t^3}{6}\right) + \mathcal{O}(\epsilon^2) \end{aligned}$$

A small Example

Alternatively, look for series solution $u = u_0 + \epsilon u_1 + \dots$

$$\frac{d}{dt}(u_0 + \epsilon u_1 + \dots) + \epsilon(u_0 + \epsilon u_1 + \dots) = t$$

Boundary condition becomes $u_0(0) + \epsilon u_1(0) + \dots = 1$

Equating coefficients of ϵ

$$u_0(0) = 1 \quad u_1(0) = u_n(0) = 0$$

Equating coefficients in governing equation and applying BCs

$$\epsilon^0 : \quad \frac{du_0}{dt} = t \quad \Rightarrow \quad u_0 = \frac{t^2}{2} + 1$$

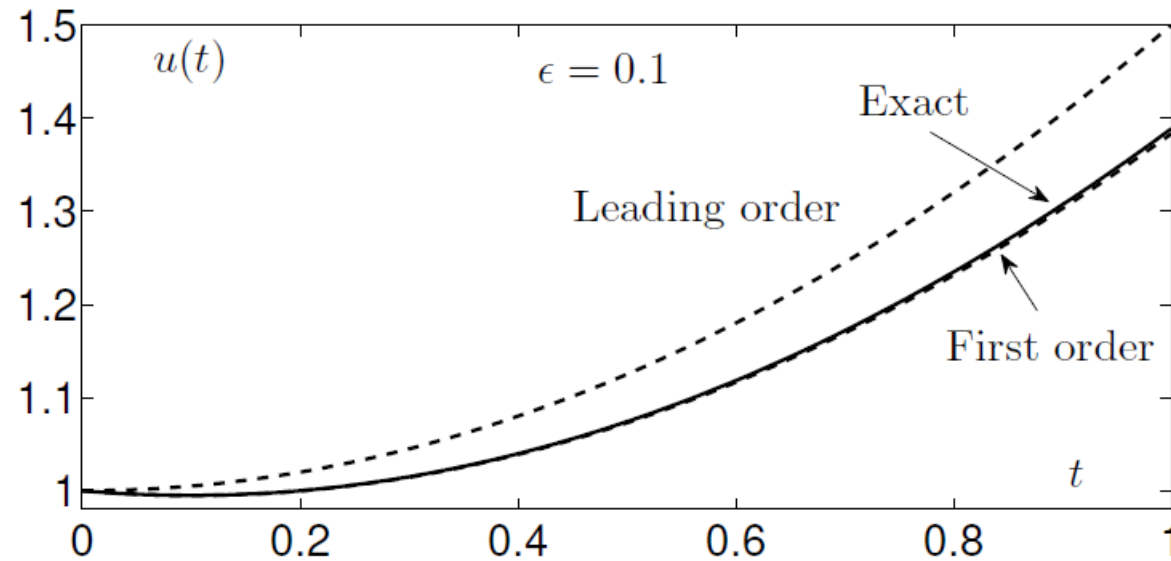
$$\epsilon^1 : \quad \frac{du_1}{dt} + u_0 = 0 \quad \Rightarrow \quad u_1 = -t - \frac{t^3}{6}$$

We can exploit this method for heat flow, phase change

etc Note, series of simpler problems

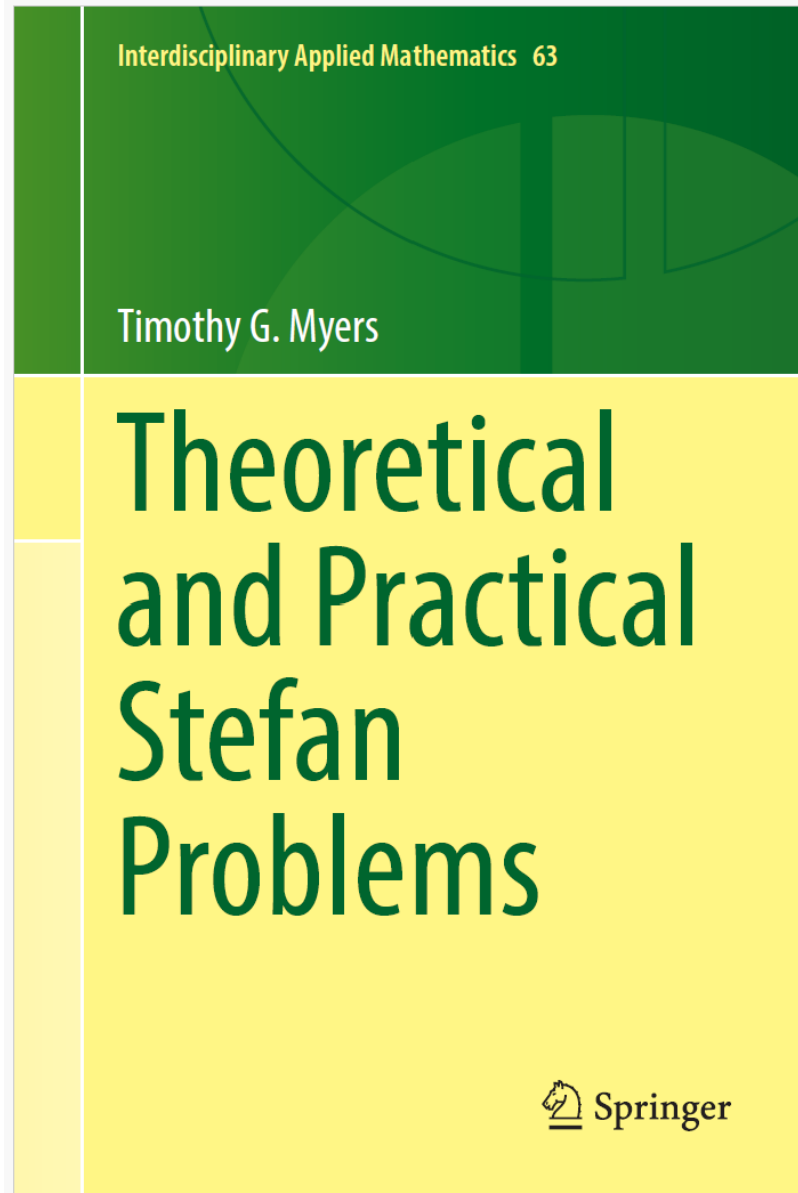
As expected (hoped?) asymptotic solution = expansion of exact solution

$$u = 1 + \frac{t^2}{2} - \epsilon \left(t + \frac{t^3}{6} \right) + \mathcal{O}(\epsilon^2)$$



Since $\epsilon = 0.1$ expect errors of 10% at leading order (**in practice usually less**). At first order we neglect terms of $\mathcal{O}(\epsilon^2)$ and therefore expect errors of 1%. From the figure it is clear that the first order solution is more accurate than the leading order, and the accuracy could be improved by looking for higher order corrections. Accuracy will improve as ϵ decreases.

We are finally ready for ...



The New York Times
bestseller

<https://doi.org/10.1007/978-3-032-04826-4>

The Stefan problem (finally!!!)



- 1 gravity and viscous effects are negligible;
- 2 the thermophysical properties, such as specific heat and thermal conductivity, are constant within each phase (often they are set to the same value in both phases);
- 3 the phase change temperature is constant;
- 4 the density, ρ , is constant and the same in both phases;
- 5 the interface between the two phases is sharp (*i.e* of zero width).

The Stefan problem (finally!!!)



$$\rho_s c_s \frac{\partial T}{\partial t} = k_s \frac{\partial^2 T}{\partial x^2}, \quad 0 < x < s(t),$$

Heat equations in each domain

$$\rho_l c_l \frac{\partial \theta}{\partial t} = k_l \frac{\partial^2 \theta}{\partial x^2}, \quad s(t) < x,$$

$$\rho_s L_m \frac{ds}{dt} = k_s \frac{\partial T}{\partial x} \Big|_{x=s} - k_l \frac{\partial \theta}{\partial x} \Big|_{x=s},$$

The Stefan condition (a conservation equation) -
Energy balance to determine
moving boundary position

Same boundary conditions but we have introduced a new derivative –

One domain may not exist at $t=0$

Boundary temperature is phase change temperature (provides a BC for both domains)

More general form

$$\rho_s c_s \frac{\partial T}{\partial t} = k_s \nabla^2 T ,$$

$$\rho_l c_l \frac{\partial \theta}{\partial t} = k_l \nabla^2 \theta ,$$

$$\rho_s L_m \frac{dR}{dt} = (k_s \nabla T - k_l \nabla \theta)|_R \cdot \hat{\mathbf{n}} ,$$

With flowing liquid

$$\rho_l c_l \left(\frac{\partial \theta}{\partial t} + \mathbf{u} \cdot \nabla \theta \right) = k_l \nabla^2 \theta ,$$

Non-dimensionalisation

$$\hat{x} = \frac{x}{L}, \quad \hat{s} = \frac{s}{L}, \quad \hat{t} = \frac{t}{\tau}, \quad \hat{\theta} = \frac{\theta - T_m}{\Delta\theta}, \quad \hat{T} = \frac{T - T_m}{\Delta T},$$

Solid heat equation ..

$$\frac{\rho_s c_s \Delta T}{\tau} \frac{\partial \hat{T}}{\partial \hat{t}} = \frac{k_s \Delta T}{L^2} \frac{\partial^2 \hat{T}}{\partial \hat{x}^2}.$$

Choose time-scale with
solid heat flow

$$\frac{\partial \hat{T}}{\partial \hat{t}} = \frac{\partial^2 \hat{T}}{\partial \hat{x}^2}$$

$$\frac{\partial \hat{\theta}}{\partial \hat{t}} = \alpha \frac{\partial^2 \hat{\theta}}{\partial \hat{x}^2}$$

$$\alpha = \alpha_l / \alpha_s = k_l \rho_s c_s / (k_s \rho_l c_l)$$

$$k = k_l \Delta\theta / (k_s \Delta T)$$

Stefan condition

$$\rho_s L_m \frac{L}{\tau} \frac{d\hat{s}}{d\hat{t}} = \frac{k_s \Delta T}{L} \frac{\partial \hat{T}}{\partial \hat{x}} - \frac{k_l \Delta\theta}{L} \frac{\partial \hat{\theta}}{\partial \hat{x}}$$

$$\beta \frac{d\hat{s}}{d\hat{t}} = \frac{\partial \hat{T}}{\partial \hat{x}} - k \frac{\partial \hat{\theta}}{\partial \hat{x}}$$

Stefan (or inverse) number

$$\beta = \frac{L_m}{c_s \Delta T}$$

Time-scale ...

Substance	T_m (K)	L_m (J/Kg)	c_l, c_s (J/Kg·K)	ρ_l, ρ_s (kg/m ³)	k_l, k_s (W/m·K)	σ_{sl} (N/m)
Water	273.15	3.34×10^5	4181, 2050	1000, 920	0.55, 2.20	0.03
Gold	1337	6.37×10^4	163, 129	17,300, 19,300	106, 317	0.27
Lead	600	2.30×10^4	148, 128	10,700, 11,300	16,35	0.05
Tin	505	5.85×10^4	268, 230	6980, 7180	30, 67	0.064
Copper	136	2.09×10^5	530, 385	8020, 8960	157.2, 413	0.182
Salol	316	9×10^4	1580, -	1180, 1250	0.18, -	-
n-Octadecane	301.3	2.435×10^5	2161, 2151	771, 930	0.15, 0.372	-

Stefan number is generally large $\beta \gg 1$

$L_m/c_s \approx 162, 493, 179, 254\text{K}$ for water, gold, lead and tin

For water, with temperature scale of 5K $\beta \approx 30$

Need 162K for $\beta = 1$

Time-scale?

$$\frac{d\hat{s}}{d\hat{t}} = \frac{1}{\beta} \left(\frac{\partial \hat{T}}{\partial \hat{x}} - k \frac{\partial \hat{\theta}}{\partial \hat{x}} \right) \quad \text{As } \beta \rightarrow \infty \quad \hat{s}_{\hat{t}} \rightarrow 0$$

So, the physically interesting limit of large β suggests the speed of the front becomes negligible – we miss the very effect we are interested in!

Why?

Heat Flow is generally a lot faster than phase change

$$\rho_s L_m \frac{L}{\tau} \frac{d\hat{s}}{d\hat{t}} = \frac{k_s \Delta T}{L} \frac{\partial \hat{T}}{\partial \hat{x}} - \frac{k_l \Delta \theta}{L} \frac{\partial \hat{\theta}}{\partial \hat{x}}$$

Don't choose the time-scale

$$\Rightarrow \beta = \frac{\rho_s L_m L^2}{\tau k_s \Delta T}$$

Time-scale?

$$\rho_s L_m \frac{L}{\tau} \frac{d\hat{s}}{d\hat{t}} = \frac{k_s \Delta T}{L} \frac{\partial \hat{T}}{\partial \hat{x}} - \frac{k_l \Delta \theta}{L} \frac{\partial \hat{\theta}}{\partial \hat{x}} \qquad \beta = \frac{\rho_s L_m L^2}{\tau k_s \Delta T}$$
$$\tau = \rho_s L_m L^2 / (k_s \Delta T)$$

Stefan condition balanced, heat equations contain a **small parameter**

$$\frac{d\hat{s}}{d\hat{t}} = \frac{\partial \hat{T}}{\partial \hat{x}} - k \frac{\partial \hat{\theta}}{\partial \hat{x}} \qquad \frac{1}{\beta} \frac{\partial \hat{T}}{\partial \hat{t}} = \frac{\partial^2 \hat{T}}{\partial \hat{x}^2}, \qquad \frac{1}{\beta} \frac{\partial \hat{\theta}}{\partial \hat{t}} = \alpha \frac{\partial^2 \hat{\theta}}{\partial \hat{x}^2}$$

In the limit $\beta \rightarrow \infty$ this indicates

$$\frac{\partial^2 \hat{T}}{\partial \hat{x}^2}, \frac{\partial^2 \hat{\theta}}{\partial \hat{x}^2} \rightarrow 0$$

Steady-state?

In the limit $\beta \rightarrow \infty$ this indicates

$$\frac{\partial^2 \hat{T}}{\partial \hat{x}^2}, \frac{\partial^2 \hat{\theta}}{\partial \hat{x}^2} \rightarrow 0$$

So, in the limit of large β the temperature tends to the steady-state?

Not quite – the BC $T=0$ ($T=T_m$) is applied at a moving boundary

$$T(s(t), t) = 0$$

This is termed a pseudo- or quasi- steady state and the distinction from a steady-state is

- a) frequently misunderstood,
- b) crucial to the solution

Remember this!

Exact solutions

There exist many solutions but hardly any are of practical use – most are chosen to satisfy the governing equations (recall we can generate solutions to the heat equation) and then seeing what BCs they satisfy .

We will now examine a few practical (but also restrictive) exact solutions

Exact solutions

Back to the first problem ...



$$\frac{\partial T}{\partial t} = \frac{\partial^2 T}{\partial x^2}$$

$$0 < x < s$$

$$\frac{\partial \theta}{\partial t} = \alpha \frac{\partial^2 \theta}{\partial x^2}$$

$$x > s$$

$$\beta \frac{ds}{dt} = \frac{\partial T}{\partial x} - k \frac{\partial \theta}{\partial x}$$

$$x = s,$$

$$\Delta T = \Delta \theta = T_i - T_m$$

$$T(0, t) = T_0, \quad T(s, t) = \theta(s, t) = 0, \quad \theta|_{x \rightarrow \infty} = \theta(x, 0) = 1$$

Neumann's solution



Work with $\eta = x/\sqrt{t}$ $T(x,t) = F_1(\eta)$, $\theta(x,t) = F_2(\eta)$,

$$\frac{\partial T}{\partial t} = \frac{dF_1}{d\eta} \frac{\partial \eta}{\partial t} = \frac{dF_1}{d\eta} \left(-\frac{x}{2t^{3/2}} \right) = \frac{dF_1}{d\eta} \left(-\frac{\eta}{2t} \right),$$

$$\frac{\partial T}{\partial x} = \frac{dF_1}{d\eta} \frac{\partial \eta}{\partial x} = \frac{1}{\sqrt{t}} \frac{dF_1}{d\eta},$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{1}{\sqrt{t}} \frac{dF_1}{d\eta} \right) = \frac{1}{\sqrt{t}} \frac{\partial \eta}{\partial x} \frac{\partial}{\partial \eta} \left(\frac{dF_1}{d\eta} \right) = \frac{1}{t} \frac{d^2 F_1}{d\eta^2}$$

Neumann's solution

Solid heat equation $-\left(\frac{\eta}{2}\right) \frac{dF_1}{d\eta} = \frac{d^2 F_1}{d\eta^2}$

This must be independent of x and t

$$\partial F_1 / \partial \eta = G_1 \quad \frac{1}{G_1} \frac{dG_1}{d\eta} = -\frac{\eta}{2} \Rightarrow G_1 = \frac{dF_1}{d\eta} = A' e^{-\eta^2/4}$$

$$x = 0 \text{ corresponds to } \eta = 0 \Rightarrow F_1(0) = T_0$$

$$F_1 = T_0 + A \operatorname{erf} \left(\frac{\eta}{2} \right) \Rightarrow T = T_0 + A \operatorname{erf} \left(\frac{x}{2\sqrt{t}} \right)$$

$$\text{Similar} \quad \theta = 1 - B \operatorname{erfc} \left(\frac{x}{2\sqrt{\alpha t}} \right)$$

Neumann's solution

At the moving interface, $x = s(t)$ $T = \theta = 0$

$$0 = T_0 + A \operatorname{erf}\left(\frac{s}{2\sqrt{t}}\right) = 1 - B \operatorname{erfc}\left(\frac{s}{2\sqrt{\alpha t}}\right)$$

But A, B are constant, how???

The solution can only be valid if $s \propto \sqrt{t}$

Setting $s = 2\lambda\sqrt{t}$

determines $A = -\frac{T_0}{\operatorname{erf}\lambda}$ $B = \frac{1}{\operatorname{erfc}(\lambda/\sqrt{\alpha})}$

Neumann's solution

The temperatures are then

$$F_1 = T_0 \left(1 - \frac{\operatorname{erf}(\eta/2)}{\operatorname{erf}(\lambda)} \right) \Rightarrow T = T_0 \left(1 - \frac{\operatorname{erf}(x/2\sqrt{t})}{\operatorname{erf}(\lambda)} \right),$$

$$F_2 = 1 - \frac{\operatorname{erfc}(\eta/(2\sqrt{\alpha}))}{\operatorname{erfc}(\lambda/\sqrt{\alpha})} \Rightarrow \theta = 1 - \frac{\operatorname{erfc}(x/(2\sqrt{\alpha t}))}{\operatorname{erfc}(\lambda/\sqrt{\alpha})},$$

There remains one unknown ... λ

Neumann's solution

Haven't used the Stefan condition

$$\beta \frac{ds}{dt} = \frac{\partial T}{\partial x} - k \frac{\partial \theta}{\partial x}$$

Noting that

$$\frac{\partial}{\partial x} \left(\operatorname{erf} \left(\frac{x}{2\sqrt{t}} \right) \right) = -\frac{\partial}{\partial x} \left(\operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right) \right) = \frac{\exp(-x^2/(4t))}{\sqrt{\pi t}}$$

The Stefan condition is applied at $x = s = 2\lambda\sqrt{t}$.

This leads to the Transcendental Equation

$$\beta\lambda = -\frac{1}{\sqrt{\pi}} \left[\frac{T_0 e^{-\lambda^2}}{\operatorname{erf}(\lambda)} + \frac{k e^{-\lambda^2/\alpha}}{\sqrt{\alpha} \operatorname{erfc}(\lambda/\sqrt{\alpha})} \right]$$

Neumann's solution

$$\beta\lambda = -\frac{1}{\sqrt{\pi}} \left[\frac{T_0 e^{-\lambda^2}}{\operatorname{erf}(\lambda)} + \frac{k e^{-\lambda^2/\alpha}}{\sqrt{\alpha} \operatorname{erfc}(\lambda/\sqrt{\alpha})} \right]$$

So, this needs solving numerically but, it is a single equation – we have reduced a system of 2 2nd order PDEs defined over a moving domain, coupled to a first order ODE, to solving a single equation. This is a major simplification.

NB, although time appears in the derivatives, it has cancelled out of the equation for λ . This is necessary for a constant value and demonstrates that the solution form is feasible.

Neumann's solution

Freezing of water with initial temperature 293K, wall temperature 263K
 $\beta=8$

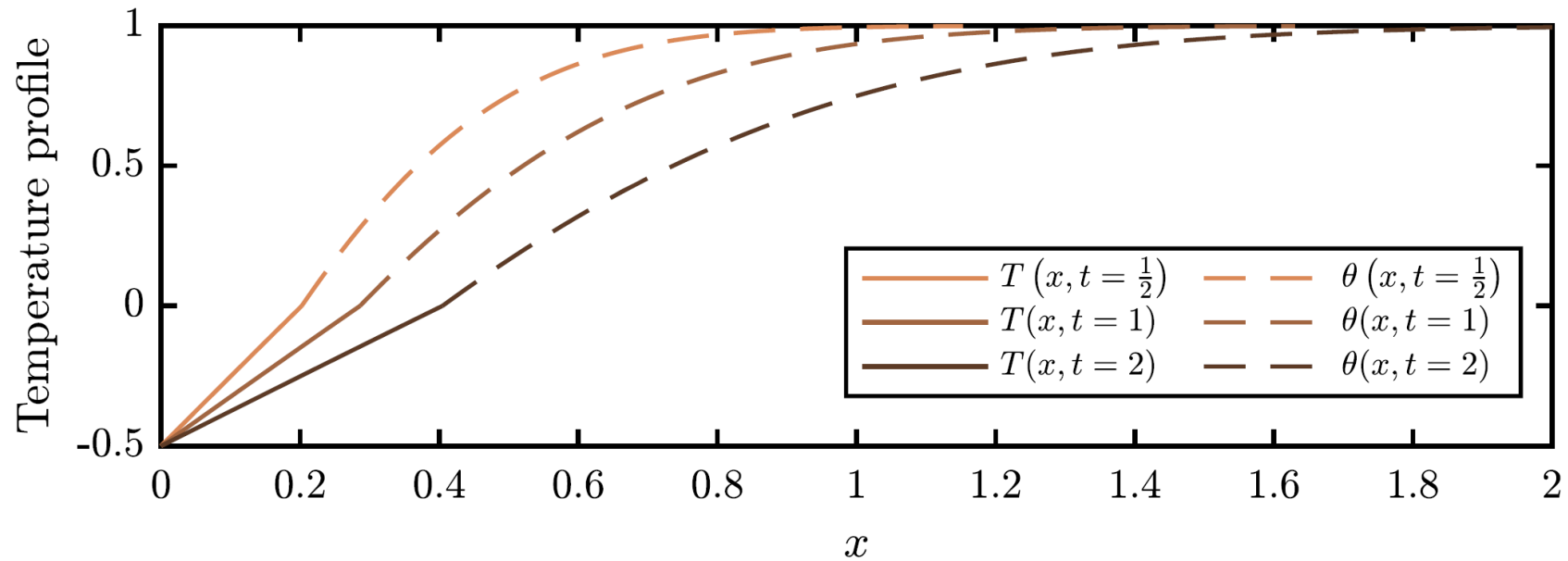


Fig. 2.1 Temperature profiles for two-phase solidification at times $t = 0.5, 1, 2$, with $T_0 = -0.5$, $k = 0.227$, $\alpha = 0.113$, $\beta = 8$ (which, from Eq. (2.12), determines $\lambda = 0.143$) solid lines represent $T(x, t)$, and dashed lines represent $\theta(x, t)$. As time increases, the interface moves between $s = 0.202, 0.286$, and 0.405



That's the end of Lecture 1!!!